Partitions of Equiangular Tight Frames

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Outline

1. Preliminaries
   - Motivation
   - The MSS Theorem & Partitions
   - Definitions
   - Construction of ETFs
   - Norm of the R-Matrix
2. Diagonal Partitions
   - Diagonal Sub-Blocks
   - Subset Norms
   - Diagonal Sub-block Norms and MSS
   - Frame Partition Algorithm

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Kadison-Singer (KS) and Equivalent Problems

- KS conjecture probed the existence of unique extensions of *linear functionals* on C^*-algebras.
- Equivalent to the *paving* conjecture.
- Equivalent to the *discrepancy* statement proven by Marcus-Spielman-Srivastava (MSS Theorem) in 2013.
- The MSS Theorem helps prove KS conjecture is true.
- We explore an open problem involving tight frames and we consider a special class of *complex* Equiangular Tight Frames (ETFs).
- The MSS Theorem states the existence of a partition of a set of vectors; however, no algorithm exists for finding such a partition.

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MSS Theorem

Theorem (Marcus-Spielman-Srivastava [6])

Let r be a positive integer and $\mathcal{V} = \{v_1, \ldots, v_n\}$ be a set of vectors in $\mathbb{C}^m$ satisfying

$$\sum_{j=1}^{n} v_j \otimes v_j = \mathbf{I}, \tag{1}$$

where $\|v_j\|^2 \leq \delta$ for all j. Then there exists a partition $\mathcal{P} = \{\mathcal{S}_1, \ldots, \mathcal{S}_r\}$ of $[n] = \{1, \ldots, n\}$ such that the subset norms satisfy

$$\left\| \sum_{j \in \mathcal{S}_k} v_j \otimes v_j \right\| \leq \left(\frac{1}{\sqrt{r}} + \sqrt{\delta} \right)^2 \tag{2}$$

for all $k \in \{1, \ldots, r\}$.

Results

1. We find a linear algorithm to partition Equiangular Tight Frames (ETFs), generated by skew symmetric conference matrices.
2. Our linear-algorithm generates a partition $\mathcal{P} = \{\mathcal{S}_1, \ldots, \mathcal{S}_r\}$ such that the bound on the subset norms is *sharper* than (2) with $\delta = 1/2$.

$$\left\| \sum_{j \in \mathcal{S}_h} f_j \otimes f_j \right\| \leq \frac{1}{2} + \frac{1}{\sqrt{2r}} \tag{3}$$

3. Partitions with subset sizes of 2 & 3 will always be MSS partitions.

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Some Definitions

Definition (Frame [1, 2, 3, 5])

A frame for a Hilbert space $\mathbb{H}$ is a sequence of vectors $\mathcal{F} = \{f_j\} \subset \mathbb{H}$ for which there exists constants $0 < A \leq B < \infty$ such that for every vector $x \in \mathbb{H}$,

$$A\|x\|^2 \leq \sum_j |\langle x, f_j \rangle|^2 \leq B\|x\|^2. \tag{4}$$

- A frame is a relaxation of a basis.
- Linear independence not required.
- A frame spans its space $\mathbb{H}$.

Some Definitions

Definition (ETF[5, 7])

A family of vectors $\mathcal{F} = \{f_j\}$ in $\mathbb{H}$ is an *equiangular-tight frame* if

1. $\|f_j\|_2 = 1$ for all j,
2. $|\langle f_j, f_k \rangle| = c$ for all $j \neq k$ and some constant c,
3. and $\sum f_j \otimes f_j = a\mathbf{I}$ for $a > 0$.

Definition (Subset Norm)

Let $\mathcal{F} = \{f_1, \ldots, f_n\}$ be a set of vectors and $\mathcal{S}$ a subset of $[n] = \{1, \ldots n\}$. We define the *subset outer product* of $\mathcal{S}$ to be matrix

$$\mathbf{F}_{\mathcal{S}} = \sum_{j \in \mathcal{S}} f_j \otimes f_j$$

and the *subset norm* of $\mathcal{S}$ to be the induced matrix (spectral) norm $\|\mathbf{F}_{\mathcal{S}}\|$.

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Construction of ETFs

Follow a construction by Strohmer & Heath [8].

Corollary ([8])

Let $m, n \in \mathbb{N}$ with $n \geq m$. Assume $\mathbf{R}$ is a Hermitian $n \times n$ matrix in $\mathbb{H}$ with entries $\mathbf{R}_{k,k} = 1$ and

$$\mathbf{R}_{k,l} = \begin{cases} \pm\sqrt{\frac{n-m}{m(n-1)}} & \text{if } \mathbb{H} = \mathbb{R}, \\ \pm i\sqrt{\frac{n-m}{m(n-1)}} & \text{if } \mathbb{H} = \mathbb{C}, \end{cases} \tag{5}$$

for $k, l = 1, \ldots, n$ with $k \neq l$. If the eigenvalues $\lambda_1, \ldots, \lambda_n$ of $\mathbf{R}$ are such that $\lambda_1 = \ldots = \lambda_m = \frac{n}{m}$ and $\lambda_{m+1} = \ldots = \lambda_n = 0$, then there exists a frame $\{f_k\}_{k=1}^n$ in $\mathbb{H}^m$ that achieves the Welch bound, i.e., $\mathcal{F} = \{f_k\}_{k=1}^n$ is a unit-norm ETF.

Construction of ETFs

We review Strohmer and Heath's construction of $\mathbf{R}$ via conference matrices by assuming that it takes the form

$$\mathbf{R} = \mathbf{I} + i\alpha\mathbf{C}, \tag{6}$$

where α is a constant and $\mathbf{C}$ is a conference matrix as defined by Geothals and Seidel [4]:

Definition

A matrix $\mathbf{C}$ of order n with diagonal elements 0 and off diagonal elements ± 1 satisfying

$$\mathbf{C}\mathbf{C}^T = (n-1)\mathbf{I} \tag{7}$$

is said to be a conference matrix.

Construction of ETFs

We recursively construct *skew-symmetric* conference matrices as follows:

$$\mathbf{C}(1) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

and define $\mathbf{C}(k)$ recursively by

$$\mathbf{C}(k) = \begin{bmatrix} \mathbf{C}(k-1) & \mathbf{C}(k-1) - \mathbf{I}_{2^{k-1}} \\ \mathbf{C}(k-1) + \mathbf{I}_{2^{k-1}} & -\mathbf{C}(k-1) \end{bmatrix}. \tag{8}$$

Construction of ETFs

Construct unit-norm ETFs from (6) by using the spectral decomposition

$$\mathbf{R} = \mathbf{W}\mathbf{\Lambda}\mathbf{W}^*$$

Corollary

Let $\mathbf{R}$ *be defined as in (6). Then the set of vectors* $\mathcal{F} = \{f_l\}_{l=1}^n$, *where*

$$f_j = \{\mathbf{W}_{j,l}\}_{l=1}^m \in \mathbb{H}^m \tag{9}$$

for $j \in \{1, \ldots, n\}$ *and* $l \in \{1, \ldots, m\}$, *is an ETF that satisfies the hypothesis of the MSS theorem, namely,* $\mathbf{F} = \mathbf{I}_m$.

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Norm of the **R**-Matrix

Theorem

The norm of the matrix $\mathbf{R} = \mathbf{I} + i\alpha\mathbf{C}$*, where* $\mathbf{C}$ *is a* $n \times n$ *skew symmetric conference matrix of order* $n = 2^k$ *and* α *is a real constant, is given by*

$$\|\mathbf{R}\| = 1 + |\alpha|\sqrt{n-1}.$$

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Finding Diagonal Sub-Blocks

We use diagonal sub-blocks because they correspond to lower dimension skew-symmetric conference matrices.

By the recursive construction of our skew-symmetric conference matrices we have

$$\mathbf{R}(k) = \begin{bmatrix} \mathbf{R}_{2,1}(k) & \star & \star & \star \\ \star & \mathbf{R}_{2,2}(k) & \star & \star \\ \star & \star & \mathbf{R}_{2,3}(k) & \star \\ \star & \star & \star & \mathbf{R}_{2,4}(k) \end{bmatrix},$$

where $\mathbf{R}_{2,q}(k) = \mathbf{R}(k-2)$ for $q = 1, 4$ and $\mathbf{R}_{2,q}(k) = \overline{\mathbf{R}}(k-2)$ for $q = 2, 3$.

$\mathbf{R}_{2,q}(k)$ is a *recursive diagonal sub-block* of $\mathbf{R}(k)$ at depth 2.

Norms of the Diagonal Sub-Blocks

Theorem

Let $\mathbf{R}(k) = \mathbf{I} + i\alpha\mathbf{C}(k)$ *where* $\mathbf{C}(k)$ *is a recursively defined skew-symmetric conference matrix of order* 2^k *and* $\alpha = 1/\sqrt{2^k - 1}$. *Then the norm of each recursive diagonal sub-block* $\mathbf{R}_{d,q}(k)$ *at depth* d *is*

$$\|\mathbf{R}_{d,q}(k)\| = 1 + \alpha\sqrt{2^{k-d} - 1}. \tag{10}$$

Theorem

Let $\mathbf{R}(k) = \mathbf{I} + i\alpha\mathbf{C}(k)$ *where* $\mathbf{C}(k)$ *is a skew-symmetric conference matrix of order* 2^k *defined by (8). Then the entries of the* q*-th diagonal sub-block* $\mathbf{R}_{d,q}(k)$ *at recursive depth* d *are given by* $(\mathbf{R}_{d,q}(k))_{g,h} = f_h^* f_g$, *where* $(g, h) \in \mathcal{S}_{d,q} \times \mathcal{S}_{d,q}$ *and* $\mathcal{S}_{d,q}$ *is defined by*

$$\mathcal{S}_{d,q} = \{(q-1) \cdot 2^{k-d} + 1, \ldots, q \cdot 2^{k-d}\}, \tag{11}$$

for $q \in \{1, \ldots, 2^d\}$ *and* $d \in \{0, \ldots, k-1\}$.

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Subset Norms Correspond to Sub-Matrix Norms

Suppose $\mathcal{S} = \{j_r\}_{r=1}^q \subset \mathcal{I}$ is a q-element subset of $\mathcal{I}$ and $\mathcal{F}_\mathcal{S} = \{f_{j_r}\}_{r=1}^q$ a linearly independent set of frame vectors. As before, we define

$$\mathbf{F}_\mathcal{S} = \sum_{i \in \mathcal{S}} f_i \otimes f_i = \sum_{r=1}^{q} f_{j_r} f_{j_r}^*.$$

To find the norm $\|\mathbf{F}_\mathcal{S}\|$, we shall solve the eigenvalue problem $\mathbf{F}_\mathcal{S} v = \lambda v$, where $v \in \mathbb{C}^m$.
Assume $\lambda \neq 0$. Then since $\mathbf{F}_\mathcal{S} v \in \operatorname{span}(\mathcal{F}_\mathcal{S})$ and $\mathcal{F}_\mathcal{S}$ is a linearly independent set, we may assume that the $v \in \operatorname{span}(\mathcal{F}_\mathcal{S})$, i.e.,

$$v = \sum_{r=1}^{q} c_{j_r} f_{j_r}$$

for some set of coefficients $\{c_1, \ldots, c_q\}$.

Subset Norms Correspond to Sub-Matrix Norms

We obtain a set of equations which can be expressed as the following matrix equation:

$$\begin{bmatrix} f_{j_1}^* f_{j_1} & f_{j_1}^* f_{j_2} & f_{j_1}^* f_{j_3} & \cdots & f_{j_1}^* f_{j_q} \\ f_{j_2}^* f_{j_1} & f_{j_2}^* f_{j_2} & f_{j_2}^* f_{j_3} & \cdots & f_{j_2}^* f_{j_q} \\ f_{j_3}^* f_{j_1} & f_{j_3}^* f_{j_2} & f_{j_3}^* f_{j_3} & \cdots & f_{j_3}^* f_{j_q} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ f_{j_q}^* f_{j_1} & f_{j_q}^* f_{j_2} & f_{j_q}^* f_{j_3} & \cdots & f_{j_q}^* f_{j_q} \end{bmatrix} \begin{bmatrix} c_{j_1} \\ c_{j_2} \\ c_{j_3} \\ \vdots \\ c_{j_q} \end{bmatrix} = \lambda \begin{bmatrix} c_{j_1} \\ c_{j_2} \\ c_{j_3} \\ \vdots \\ c_{j_q} \end{bmatrix}. \tag{12}$$

The matrix on the left is a sub-matrix of $\mathbf{R}$; therefore, this is equivalent to solving

$$\frac{m}{n}\mathbf{R}_{\mathcal{S}} c_{\mathcal{S}} = \lambda c_{\mathcal{S}}. \tag{13}$$

Subset Norms Correspond to Sub-Matrix Norms

Theorem

Let $\mathcal{F} = \{f_j\}_{j=1}^n$ be an ETF defined by (9) and $\mathcal{S} \subset \mathcal{I}$ an arbitrary subset with $\mathcal{F}_\mathcal{S}$ a linearly independent set of frame vectors. Then

$$\|\mathbf{F}_\mathcal{S}\| = \frac{m}{n}\|\mathbf{R}_\mathcal{S}\|,$$

where the entries of $\mathbf{R}_\mathcal{S}$ correspond to the entries $(g, h) \in \mathcal{S} \times \mathcal{S}$ of $\mathbf{R}$.

We couple this with our construction of $\mathbf{R}$ and taking subsets that correspond to the recursive diagonal sub-blocks we obtain the following refinement to the norm:

Diagonal Sub-Block Norm

Theorem

Let $\mathcal{F} = \{f_j\}_{j=1}^{2^k}$ be an ETF defined by (9), where $\mathbf{R}(k) = \mathbf{I} + i\alpha\mathbf{C}(k)$ with $\mathbf{C}(k)$ is a skew-symmetric conference matrix of order $n = 2m = 2^k$ and $\alpha = 1/\sqrt{2^k - 1}$. Then the norm of $\mathbf{F}_{\mathcal{S}_{d,q}} = \sum_{j \in \mathcal{S}_{d,q}} f_j \otimes f_j$, where $\mathcal{S}_{d,q} \subset \mathcal{I} = [2^k]$ is a subset corresponding to q-th recursive diagonal sub-block $\mathbf{R}_{d,q}(k)$ at depth d of $\mathbf{R}(k)$, is given by

$$\|\mathbf{F}_{\mathcal{S}_{d,q}}\| = \frac{1}{2} + \frac{1}{2}\frac{\sqrt{2^{k-d} - 1}}{\sqrt{2^k - 1}} = \frac{1}{2}\|\mathbf{R}_{d,q}(k)\|, \tag{14}$$

where $d \in \{0, \ldots, k-1\}$.

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Finding the Sharper Bound

We are concerned with finding the norm corresponding to the *largest* subset.
Observe that $|\mathcal{S}_{max}| = 2^{k-d_{min}}$, where $d_{min} = \min\{d_1, \ldots, d_r\}$. It follows that

$$\|\mathbf{F}_{\mathcal{S}_{max}}\| = \frac{1}{2} + \frac{1}{2}\frac{\sqrt{2^{k-d_{min}} - 1}}{\sqrt{2^k - 1}}. \tag{15}$$

We now manipulate the right hand side of (15) to obtain

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$$\frac{1}{2} + \frac{1}{2}\frac{\sqrt{2^{k-d_{min}} - 1}}{\sqrt{2^k - 1}} \leq \frac{1}{2} + \frac{1}{\sqrt{2^{d_{min}+2}}}$$

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We now manipulate the right hand side of (15) to obtain

$$\begin{aligned}\frac{1}{2} + \frac{1}{2}\frac{\sqrt{2^{k-d_{min}} - 1}}{\sqrt{2^k - 1}} &\leq \frac{1}{2} + \frac{1}{\sqrt{2^{d_{min}+2}}} \\ &\leq \frac{1}{2} + \frac{1}{\sqrt{2r}} < \text{MSS Bound (2)}.\end{aligned}$$

Partition Subset Norm Theorem

Theorem

Let $\mathcal{F} = \{f_j\}_{j=1}^{2^k}$ be an ETF constructed from $\mathbf{R} = \mathbf{I} + i\alpha\mathbf{C}$ as described in Corollary by (9), where $\alpha = 1/\sqrt{2^k - 1}$ and $\mathbf{C}$ is a skew-symmetric conference matrix defined recursively by (8). Fix r to be a positive integer. Then for all diagonal partitions $\mathcal{P} = \{\mathcal{S}_1, \ldots, \mathcal{S}_r\}$ of the index set $\mathcal{I} = \{1, \ldots, 2^k\}$ with $|S_h| \leq 2^{k - \lfloor \log_2 r \rfloor}$ for all $h \in \{1, \ldots, r\}$, the subset norms satisfy

$$\left\| \sum_{j \in \mathcal{S}_h} f_j \otimes f_j \right\| \leq \frac{1}{2} + \frac{1}{\sqrt{2r}} \tag{16}$$

for all $h \in \{1, \ldots, r\}$.

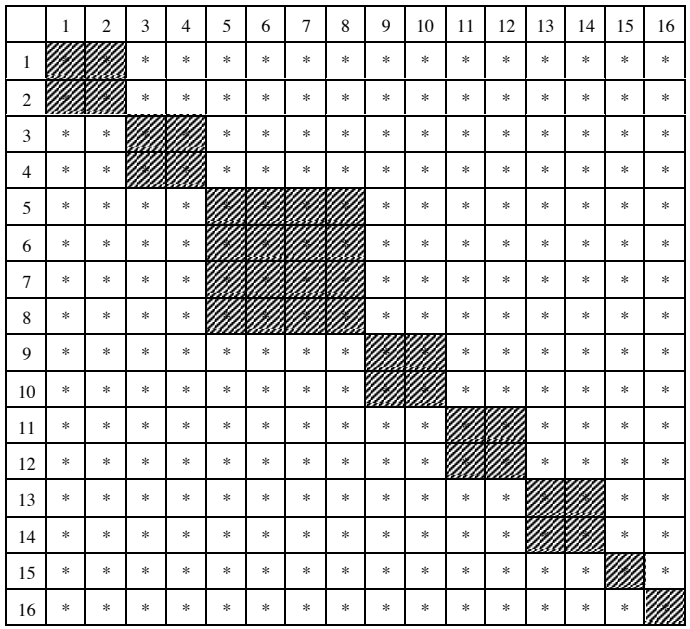
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General Partition Example

Example

- Construct a 16 vector ETF, $\mathcal{F} = \{f_j\}_{j=1}^{16}$.
- Find an $r = 8$ subset partition $\mathcal{P} = \{\mathcal{S}_1, \ldots, \mathcal{S}_8\}$ of $\mathcal{F}$.



Subset	$\|\mathbf{F}_{\mathcal{S}_h}\|$
$\mathcal{S}_1 = \{1, 2\}$	0.629099
$\mathcal{S}_2 = \{3, 4\}$	0.629099
$\mathcal{S}_3 = \{5, 6, 7, 8\}$	0.723607
$\mathcal{S}_4 = \{9, 10\}$	0.629099
$\mathcal{S}_5 = \{11, 12\}$	0.629099
$\mathcal{S}_6 = \{13, 14\}$	0.629099
$\mathcal{S}_7 = \{15\}$	0.5
$\mathcal{S}_8 = \{16\}$	0.5

Table: All norms $< \frac{1}{2} + \frac{1}{\sqrt{2r}} = 0.75$

Algorithm

Algorithm 1: Diagonal Partition Algorithm

Result: $\mathcal{I} = \{1, \ldots, 2^k\}$ is partitioned into $\mathcal{P} = \{\mathcal{S}_{d_i,q_i}\}_{i=1}^{r}$.

Given $r > 1$;

Initialize $d = \lfloor \log_2 r \rfloor$, $r' = 2^d$;

Initialize $\mathcal{P} = \emptyset$;

for $i = 1$ **to** $2(r - r')$ **do**

$\quad \mathcal{S}_{d+1,i} = \{(i-1) \cdot 2^{k-(d+1)} + 1, \ldots, i \cdot 2^{k-(d+1)}\}$;

$\quad \mathcal{P} = \mathcal{P} \cup \{\mathcal{S}_{d+1,i}\}$;

end

for $i = r - r' + 1$ **to** r' **do**

$\quad \mathcal{S}_{d,i} = \{(i-1) \cdot 2^{k-d} + 1, \ldots, i \cdot 2^{k-d}\}$;

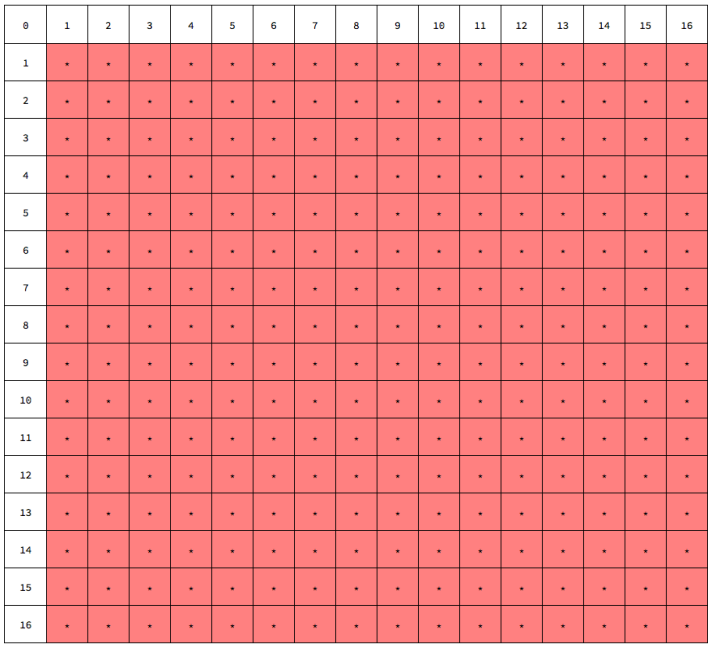
$\quad \mathcal{P} = \mathcal{P} \cup \{\mathcal{S}_{d,i}\}$;

end

Algorithm Example: $\mathcal{F} = \{f_j\}_{j=1}^{16}$

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- Construct a 16 vector ETF, $\mathcal{F} = \{f_j\}_{j=1}^{16}$.
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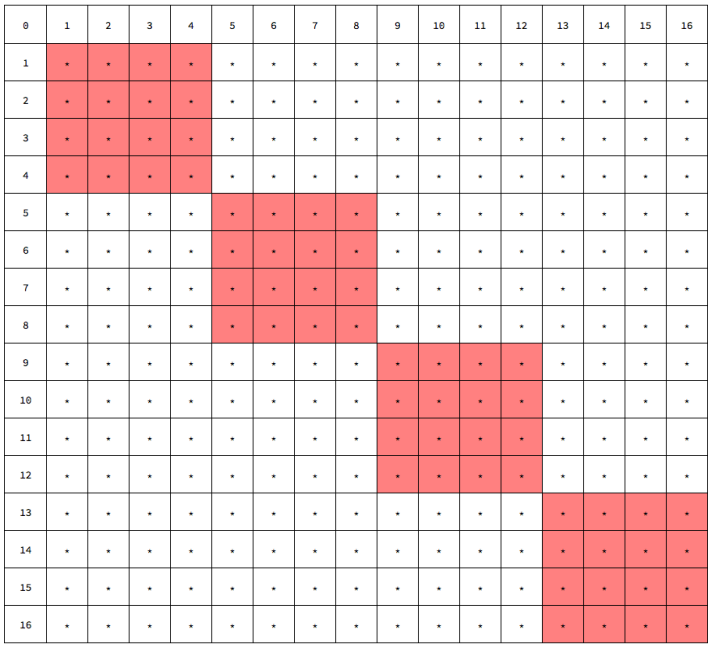


$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16\}$

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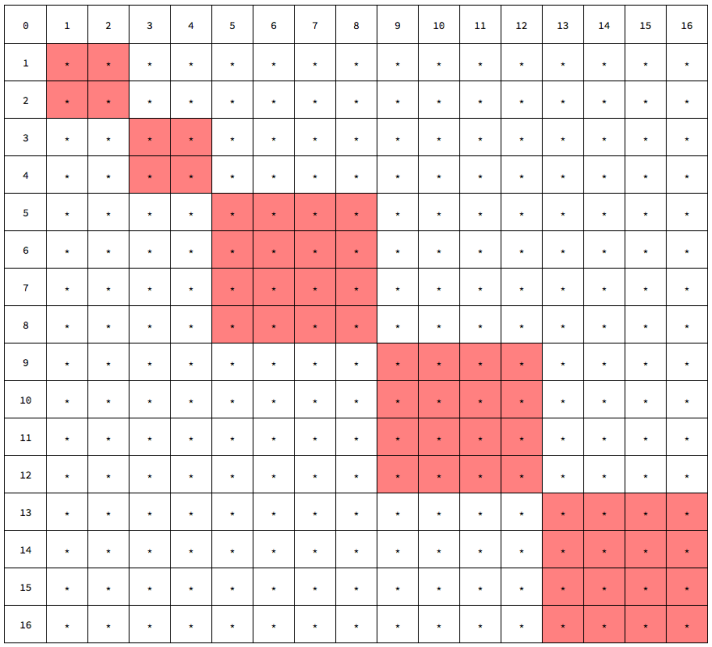
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Example

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- Our results apply to complex ETFs constructed from skew symmetric conference matrices.
- We have shown that partitions containing subsets of size 2 or 3 are MSS partitions.
- We are exploring partitions of ETFs constructed using Steiner systems.

References

- P. G. Casazza and G. Kutyniok. *Finite Frames: Theory and Applications*. New York, New York: Springer, 2013.
- O. Christensen. *An Introduction to Frames and Riesz Bases*. Boston, Massachusetts: Birkhauser, 2003.
- O. Christensen. *Frames And Bases: An Introductory Course*. Boston, Massachusetts: Birkhauser, 2008.
- J. M. Goethals and J. J. Seidel. "Orthogonal matrices with zero diagonal." In: *Canadian Journal of Mathematics* 19 (1967), pp. 1000–1010. DOI: 10.4153/CJM-1967-091-8.
- D. Han et al. *Frames for Undergraduates*. Vol. 40. Providence, Rhode Island: American Mathematical Society, 2007.
- A. Marcus, D. A. Spielman, and N. Svrivastava. "Interlacing families II: Mixed characteristic polynomials and the Kadison-Singer problem." In: *Annals of Mathematics* 182 (2015), pp. 327–350. DOI: 10.4007/annals.2015.182.1.8.
- T. Strohmer. "A note on equiangular tight frames". In: *Linear Algebra and its Applications* 429 (2008), pp. 326–330. DOI: 10.1016/j.laa.2008.02.030.
- T. Strohmer and R. Heath. "Grassmannian Frames with Applications to Coding and Communication." In: *Applied and Computational Harmonic Analysis* 41 (2003), pp. 257–275. DOI: 10.1016/S1063-5203(03)00023-X.